DATE: 2021/26
EXNO: 1

Study of various network commands used in Linux & windows.

Aim:

Study of basic network commands:-

1. arp-a: It show the IP address along with MAC address
2. hostname: Display the name of your computer
3. ipconfig/all: Display the info about TCP/IP Gateway, DNS, DHCP.
4. nbstat-a: Helps to solve problem with netBIOS
5. nbtstat: Display the variety of statistics about computer active TCP/IP
   Eg: net stat-r
6. nslookup: It is a tool used to performs DNS lookups in Linux.
7. Pathping: It is window combination of Ping and Tracert commands.
8. Ping: To test a connectivity between two nodes.

9.

Linux network commands

1. Ipconfig : Display network interface setting
2. ip -> show the address information
3. Ifconfig :

4.netstat

```
C:\Windows\system32>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    172.16.9.19:58167      lcbomp-in-f188:5228    ESTABLISHED
  TCP    172.16.9.19:58403      lcmaaa-ax-in-f5:https  ESTABLISHED
  TCP    172.16.9.19:58444      sd-in-f100:https       TIME_WAIT
  TCP    172.16.9.19:58446      4.213.25.240:https     ESTABLISHED
  TCP    172.16.9.19:58451      a104-86-188-168:https  CLOSE_WAIT
  TCP    172.16.9.19:58453      pnmaaa-ay-in-f10:https  TIME_WAIT
  TCP    172.16.9.19:58454      4.213.25.241:https     ESTABLISHED
  TCP    172.16.9.19:58455      pnmaaa-am-in-f10:https  TIME_WAIT
  TCP    172.16.9.19:58456      pnmaaa-ay-in-f10:https  ESTABLISHED
  TCP    172.16.9.19:58457      pnmaaa-bb-in-f14:https  ESTABLISHED
```

```
C:\Windows\system32>pathping google.in

Tracing route to google.in [142.250.207.68]
over a maximum of 30 hops:
  0  DESKTOP-0M11URO [172.16.9.19]
  1  172.16.8.1
  2  137.97.163.129
  3     *        *        *
Computing statistics for 50 seconds...
            Source to Here   This Node/Link
Hop  RTT    Lost/Sent = Pct  Lost/Sent = Pct  Address
  0                                           DESKTOP-0M11URO [172.16.9.19]
                                    0/ 100 =  0%   |
  1    0ms     0/ 100 =  0%     0/ 100 =  0%  172.16.8.1
                                    0/ 100 =  0%   |
  2    1ms     0/ 100 =  0%     0/ 100 =  0%  137.97.163.129

Trace complete.
```

```
:\Windows\system32>route print
===========================================================================
nterface List
 3...00 27 0e 13 ee 92 ......Realtek PCIe GbE Family Controller
 1...........................Software Loopback Interface 1
===========================================================================

Pv4 Route Table
===========================================================================
ctive Routes:
etwork Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0      172.16.8.1    172.16.9.19     291
        127.0.0.0        255.0.0.0         On-link         127.0.0.1     331
        127.0.0.1  255.255.255.255         On-link         127.0.0.1     331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1     331
       172.16.8.0    255.255.252.0         On-link      172.16.9.19     291
      172.16.9.19  255.255.255.255         On-link      172.16.9.19     291
    172.16.11.255  255.255.255.255         On-link      172.16.9.19     291
        224.0.0.0        240.0.0.0         On-link         127.0.0.1     331
        224.0.0.0        240.0.0.0         On-link      172.16.9.19     291
  255.255.255.255  255.255.255.255         On-link         127.0.0.1     331
  255.255.255.255  255.255.255.255         On-link      172.16.9.19     291
===========================================================================
ersistent Routes:
 Network Address          Netmask  Gateway Address  Metric
          0.0.0.0          0.0.0.0      172.16.8.1  Default
===========================================================================

Pv6 Route Table
===========================================================================
ctive Routes:
If Metric Network Destination      Gateway
 1    331 ::1/128                  On-link
 3    291 fe80::/64                On-link
 3    291 fe80::bc96:2f44:bfbf:5a00/128
                                   On-link
 1    331 ff00::/8                 On-link
 3    291 ff00::/8                 On-link
===========================================================================
ersistent Routes:
 None
```

5.arp -a

```
C:\Windows\system32>arp -a

Interface: 172.16.9.19 --- 0x3
  Internet Address      Physical Address      Type
  172.16.8.1            7c-5a-1c-cf-be-45     dynamic
  172.16.11.255         ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.192.152.143       01-00-5e-40-98-8f     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
```

DATE: 20-1-26 configuring an ethernet
EX NO: 2 connection by using nmcli

Aim :-

To establish a new ethernet connection by using nmcli

Procedure :-

1. List the network manager connection profiles:

#nmcli connection show

NAME	UUID	TYPE	DEVICE
Wired connection 1	a5eb64940-cc20-....	ethernet	enp1s0

2. #nmcli connection add con-name <connection-name> ifname <device-name> type ethernet

3. Rename the connection profile

#nmcli connection modify "Wired connection"

4. Display the current setting of the connection profile : #nmcli connection show.

connection.interface-name : enp1s0
connection.autoconnect : yes
ipv4.method : auto
ipv6.method : auto

5. Configure the IPv4 settings,

* To use DHCP enter:

#nmcli .connection .modify "Wired Connection 1" ipv4.method .auto skip this step if ivP4.method is already set to auto

* To set a static IPv4 address. network .mask, default gateway, DNS servers, and search domain enter:

nmcli .connection ". Wired .connection 1" ipv4.method .manual ipv4.address 192.02.1/24 ipv4.gateway 192.0.2.254 ipv4.dns 192.0.2.00 ipv4.dns-search example.com.

6. Configure the IPv6 settings :-

* To use stateless address autoconfiguration (SLAAC) enter:

#nmcli .connection modify "Wired Connection 1" ipv6.method .auto

To set a static IPv6 address network mask, default gateway

Verification

1. Display the IP settings of the NIC:

   #ip address show enp1s0

2. Display the IPv4 default gateway #ip route show

3. Display the IPv6 default gateway : #.ip -6 route show

4. Display the DNS settings : # cat /etc/resolv.conf

5. use the ping utility to verify that this host can send packets to other.

   hosts : # ping <host-name-or-IP-address>

Result :-

Hence a ethernet connection was configured using nmcli.

EX NO: 3
DATE: 24/4/24

Study of Network Cables

Aim:-

Study of different types of Network cables

a) Understand different types of network cable

1. Unshielded Twisted pair (UTP) Cable
2. Shielded Twisted Pair (STP) Cable
3. coaxial Cable
4. Fibre optic Cable

Cable Type	Category	Maximum Data Transmission	Advantages / Disadvantages	Application / use
UTP	cat-3	10 bps	Advantages + cheaper in cost + Easy to install Disadvantages + More prone to Electromagnetic	10Base-T Ether
	Cat-5	upto 100Mbps		Fast Ethernet, Gigabit eth
	cat-5e	1 Gbps		Fast ethernet
STP	cat6,6a	10Gbps	advantages + Shielded + Faster than UTP Disadvantages + Expensive + Greater installation	Gigabit Ethe 10G ethernet Widely used in data centres
STP	Cat-7	10Gbps		Gig bit Ethern 10G Ethernet (100mt)

Coaxial Cable	RG-6 RG-59 RG-11	10-100 Mbps	+ High bandwidth + Immune to interference + Low loss bandwidth + versatile Disadvantages + Limited distance + Cost + Size is bulky	speed of signal is 500 m Television network High speed internet connections
Fibre optics Cable	single mode Multi mode	100 Gbps	Advantages + High speed + High bandwidth + High security + Long distance Disadvantages + Expensive + Requires skilled installers	Maximum distance of fibre optics cable is around 100 metres

Result :-

Hence the different types of network cables are studied.

27/7/24

Ex No: 4
DATE: 8/4/26

Setting up a Local Area Network (LAN)

Aim:

To design, set up and configure a Local Area Network (LAN)

To implementation of subnetting in CISCO PACKET TRACER simulator:

Creating a network Topology:

The first step in implementing Classless IP subnetting is to create a network topology in Packet Tracer. To create a network topology in Packet Tracer, select the "New" button in the top left corner.

Adding the devices:-

Once we have created our network topology, we can add devices to it. Here, we will be adding routers, switches and PCs. To add a device, select the device from the bottom left corner and drag it onto the network topology.

Subnetting :-

To subnet the network address of 192.168.1.0/24 to provide enough space for at least 5 addresses for end devices, the switch, and the router, we can use a /27 subnet mask. This will give us 8 subnets with 30 host address each.

The IP addressing for the network shown in the topology can be as follows

Router R1:

Gigabit Ethernet 0/0: 192.168.1.1

Gigabit Ethernet 0/1: 192.168.2.1

Switch S1:

FastEthernet 0/1: 192.168.1.0/27

PC1: 192.168.1.11

PC2: 192.168.1.12

FastEthernet 0/2: 192.168.2.0/27

PC1: 192.168.2.11

PC2: 192.168.2.12

FastEthernet 0/2: 192.168.4.0/27

PC1: 192.168.4.11

PC2: 192.168.4.12

Configuring the Devices :-

Now that we have added our devices and connected them, we can start configuring them routers. Right-click on the router and select "CLI". This will open the CLI for the router.

```
# enable
# configure terminal
# Interface FastEthernet 0/0
# ip address {IP address} {subnet mask}
# no shutdown
# exit
```

Testing the network :-

Now that our network topology is configured, we can test the network: Open a command prompt on each PC and try to ping the other PC.

Result :-

The subnetting in CISCO Packet Tracer is implemented successfully

10/8/26

EX: 05

DATE

PCs using Cisco Packet Tracer

Aim:-

To design a simple topology and configure with one router, two switches and PCs using Cisco Packet Tracer.

Build the Topology

+ Add End devices: Navigate to End Devices in the bottom-left palette. Drag four PCs into the logical workspace. Place PC0 and PC1 on the left side (LAN 1).

+ Add switches: Go to network Devices > switches. Choose the Cisco 2960 switch. Drag two switches onto the workspace.

+ Cable the Devices: Select the connection icon (the lightning bolt) and choose the copper straight-Through cable.

• Connect PC0 (Fast Ethernet 0) to switch 0 (Fast Ethernet 0/1).

• Connect PC1 (Fast Ethernet 0) to switch 0 (Fast Ethernet 0/2).

• Connect PC2 (Fast Ethernet 0) to switch 1 (Fast Ethernet 0/1).

• Connect PC3 (Fast Ethernet 0) to switch 1 (Fast Ethernet 0/2).

2. Configure the PCs

+ Click on PC0 to open its configuration Panel.

+ Go to the Desktop tab and click on IP Configuration.

+ Select static and fill in the following details:-

* IPv4 address: 192.168.10.10
+ Subnet Mask: 255.255.255.0
+ Default Gateway: 192.168.10.1

• PC1 configuration:

+ IP Address: 192.168.10.11
+ Subnet Mask: 255.255.255.0
+ Default Gateway 192.168.10.1

3. Configure the Router Interfaces

Router > enable

Router # configure terminal

Router (config) # interface Gigabit Ethernet 0/0/0.

Router (config-if) # ip address 192.168.10.1

Router (config-if) # no shutdown

Router (config-if) # exit

verify connectivity.

+ click on PC0, go to the Desktop tab, and open the CP.

+ Run a command-line ping to your local gateway to confirm the local link:

Ping 192.168.10.1

+ Test inter-subnet routing across the network.

Result:-

The network topology is operational. The first individual packet of an inter-subnet ping may initially time out due to Address Resolution Protocol (ARP).

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EX NO: 6
DATE: 17/8/24

Customize Switch with Network Modules using Cisco Packet Tracer

Aim:

To customize a modular switch in Cisco Packet Tracer by installing FastEthernet and Gigabit fiber network modules, connect it to PCs and a router, and verify network connectivity.

Procedure:

* Open Cisco Packet Tracer
* From Network Devices - Switches, select PT-EMPTY and Place it on the workspace.
* Add two PCs and one Cisco 2911 Router to the workspace
* Open the switch and go to the Physical tab.
* Turn OFF the switch before installing any modules.
* From the Modules list:
* Insert PT-switch-NM-1FFE into slot 0.
* Insert another PT-switch-NM-1FFE into slot 1.
* Insert PT-switch-NM-1FGE into slot 2.

* Open the physical tab of the Cisco 2911 router and turn it OFF.

* Insert the HWIC-1GE-SFP module into an empty slot.

* Turn the router ON and wait for it to boot.

* Check the interface using :-

Router# show ip interface brief

* verify the fiber interface is up/up

* Test connectivity using:

Router# ping 192.168.10.2

Result:-

The Switch-PT-EMPTY was successfully customized with two FastEthernet modules and one Gigabit fiber module. This CISCO 2911 router was also filled with a fiber-compatible module.

17/8/26

Ex No 1 Develop a customized Ping command to test server connectivity

Aim :-

To develop a simple python program to check whether a server is reachable the ping command.

Requirement :-

Hardware :-

+ computer
+ Internet

software :-

+ Python
+ VS code
+ Command prompt terminal

Theory :-

Ping is a network tool used to check whether a server is reachable

+ It takes a website or IP address as input
+ Sends 4 ping request
+ Displays Ping result
+ Shows whether server is reachable or unreachable

Program :-

```
import subprocess
import platform
print ("=" * 50)
Print("customized ping utility")
print ("=" * 50)
host = input ("Enter website or IP address")
system = Platform.system()
if system == "window":
    command = ["Ping", "-n", "4", host]
else
    command = ["Ping", "-c", "4", host]
try:
result: subprocessa.run(command.capture-
output = True ; time = true)
Print ("\nPing result \n")
Print (result.stdout)
if result.return code == 0;
    Print ("server status : reachable")
else:
    Print ("server status: unreachable
```

EX NO: 7 Develop a Cust
to test server

Aim:-

To develop
to check whether
ping command

Requirement:

Result :-

customized ping utility

Enter website or IP address : google.com

ping reset.

Pinging google.com [142.250.190.46] with 32 bytes of reply from 142.250.190.46.

server status : reachable

EX NO:-8 : Wireshark

Aim:-

Experiments on packet capture tool: Wireshark

* To filter, capture, view, Packets in wireshark tool capture 100 packets from the ethernet.

Procedure:-

* Select local area connection in wireshark.
* To capture option.
* Select stop capture automatically after 100 packets.
* Then click start capture
* search TCP Packets in search bar.
* To see flow graph click statistics flow.
* save the packets.

1. Create a filter to display only TCP/UDP Packets.

* Select LAN connection in wireshark
* To capture option
* Select stop capture automatically after 100 packets
* Then click start capture
* save the packets

2. Create a filter to display only ARP

+ Go to capture option
+ Select stop capture automatically after taking 100 packets
+ Then click start capture
+ search ARP packets in search bar.
+ save the packets

3). Create a filter to display only HTTP.

+ Select local area connection in wireshark.
+ Go to capture option.
+ Select stop capture automatically after 100 packets
+ Then click start capture

Result :-

Wireshark packet capture tool experiment are learned and executed successfully.

Ex: 6 Error detection & correction using Hamming code.

Aim :-
Write a program to implement error detection and correction using Hamming code.

Sender :-
* Input to sender file should be a text of any length program should be convert the text to binary

Code :-

```
data = input ("Enter 4 bits")
d1, d2, d3, d4 = map(int, data)
p1 = d1 ^ d2 ^ d4
p2 = d1 ^ d3 ^ d4
p4 = d2 ^ d3 ^ d4
code = [p1, p2, d1, p4, d2, d3, d4]
print ("sender Hamming code", "".join(map(str, code)))
```

Receiver code:-

* Receive the hamming code from the sender
* Enter the received code.
* Calculate the parity check value
* Correct the error by changing the bit
* Display the corrected hamming code.

code :-

```
r = list(map(int, input("Enter received bits")))
r1 = r[0] ^ r[2] ^ [r4] ^ [r6]
r2 = r[1] ^ r[2] ^ r[5] ^ [r6]
r3 = [r3] ^ r[4] ^ r[5] ^ r[6]
erro = r1*1 + r2*2 + r4*4
if error = 0:
    Print("No Error")
else:
    Print("Enter the position:", erro)
    r[error-1] ^= 1
    Print("corrected code:", "".join(map(str, received)))
```

Input :-

```
10 11
sender : 0110011
recei  : 0010011
Error at posdion 2
```

Result :-

Error detection & Error correction using hamming code was executed successfully.

~~10.~~

EXNO:16 Design and configure a simple internetwork using a router

Aim:-

To configure and simulate point communicate b/w two PCs using router in cisco packet tracer

Network topology :-

Device	Interface	IP adrs	Subnet
PC0	FastEthernet 0	192.16.1.2	192.168.1.1
PC1	FastEthernet 0	192.168.2.2	192.168.2.1

Procedure :-

+ Open cisco packet tracer and place two PCs and one cisco router in the workspace.

+ Construct all the connections

+ Configure static IP, subnet mask and default gateway on PC0, PC1

+ Access the router CLI mode

+ Assign IP adress to both ethernet devices

Result :-

The configuration and simulation of 2 PCs communicating.

EX.NO:11 Design and configure an internetwork using wireless router.

Aim:-

Design and configure an internetwork using wireless router, DHCP server and internet cloud.

i) Open cisco packet tracer

ii) Add network devices to the workspace and change the display

iii) Configure the network devices using DHCP servies.

iv) Configure laptop using wifi connection.

v) Click on the desktop tab at the top of the laptop configuration window and select the PC wireless ion

(vi) Configure the all PCs to set IP address. subnet

Result:-

The configuration and simulation of ~~DCP~~ DHCP server.

EXP-16 Implementation of subnetting
CISCO Packet Tracer

AIM:

To design configure and implement fixed IP subnetting (IPv4) using Cisco packet tracer simulator.

Subnetting:-

Subnetting is a process of partitioning a single physical network.

Default subnet mask: 255.255.255.0
Borrowed host bits: 3 bits
New subnet mask (CIDR): 255.255.255.224
Hosts per subnet: 25.2.30

Procedure:-

1) Open Cisco packet tracer
2) Drag 2 routers switches, PC and connect all connections
3) Divide the network 192.168.10.24 into smaller subnet using IPv4
4) Configure the router.

Result:-

Thus, Subnetting was implemented

EXNO:- 12 Implementation of subnetting CISCO Packet Tracer

Aim:-

To design configure and implement classes IP subnetting (12-1) using cisco packet tracer simulator.

Subnetting:-

subnetting is a process of partitioning a single physical network.

Default subnet mask :255.255.255.0

Borrowed host bits :3 bits

New subnet mask (127): 255.255.255.224

Hosts per subnet 25.2.30

Procedure:-

1) Open cisco packet tracer

2) Drag 2 routers switches, PC and connect all connections

3) Divide the network 192.168.10.24 into smaller subnet using 127

4) Configure the route.

Result:-

Thus. Subnetting was implemented configured.

CISCO Packet Tracer

Aim:

To design configure and implement classless IP subnetting (i.e.) using Cisco packet tracer simulator.

Subnetting:-

subnetting is a process of partitioning a single physical network.

Default subnet mask: 12.55.255.0

Borrowed host bits: 3 bits

New subnet mask (127): 255.255.224

Hosts per subnet 25.2.30

Procedure:-

1) Open cisco packet tracer

2) Drag 2 routers switches, PC and connect all connections

3) Divide the network 192.168.10.24 into smaller subnet using 127

4) Configure the route.

Result:-

Thus. Subnetting was implemented configured

EXNO:- 12 Implementation of subnetting
CISCO Packet Tracer

Aim:-

To design configure and implement classes IP subnetting (12-D) using cisco packet tracer simulator.

Subnetting :-

subnetting is a process of paitioning a single physical network.

Default subnet mask :255.255.255.0

Borrowed host bits : 3 bits

New subnet mask (127): 255.255.254

Hosts per subnet 25.2.30

Procedure :-

1) Open cisco packet tracer

2) Drag 2 routers switches, PC and connect all connections

3) Divide the network 192.168.10.24 into smaller subnet using 127

4) Configure the route.

Result :-

Thus. Subnetting was implemented configured.

Ex No: 13 Implement own ping program.

Aim :-

To implement a python program to check the connectivity b/w the local system and a specific host using the ping command.

Ping Program :-

```
import os
import platform
import subprocess
def ping(host: "gmail.com", count=4)
    Param = "-n" if platform.system().lower()=="
    command = ["ping", Param, str(count), host]
try:
    output = subprocess.check_output(
        command,
        universal_newlines = True
    )
print(output
```

Output :-

```
Ping 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8 byte = 32 time = 8ms TTL: 119
Reply from 8.8.8.8 byte = 32 time = 10ms TTL=119
Reply from 8.8.8.8 byte = 32 time 12ms TTL. 119
Reply from 8.8.8.8
```